

Stokes' Theorem and the Curl of Electrostatic Field

From electrostatics, we already know that the circulation of the electric field around any closed path is zero:

$$\oint_C \vec{E} \cdot d\vec{l} = 0. \quad (1)$$

Also, since the electric field is derivable from a scalar potential,

$$\vec{E} = -\nabla\phi = -\hat{x} \frac{\partial\phi}{\partial x} - \hat{y} \frac{\partial\phi}{\partial y} - \hat{z} \frac{\partial\phi}{\partial z}. \quad (2)$$

Now let us evaluate the curl of $\vec{E}$:

$$\nabla \times \vec{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix}. \quad (3)$$

Using $\vec{E} = -\nabla\phi$, the x -component becomes

$$\begin{aligned} (\nabla \times \vec{E})_x &= \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \\ &= \frac{\partial}{\partial y} \left(-\frac{\partial\phi}{\partial z} \right) - \frac{\partial}{\partial z} \left(-\frac{\partial\phi}{\partial y} \right) \\ &= -\frac{\partial^2\phi}{\partial y \partial z} + \frac{\partial^2\phi}{\partial z \partial y} = 0. \end{aligned} \quad (4)$$

Similarly,

$$(\nabla \times \vec{E})_y = 0, \quad (\nabla \times \vec{E})_z = 0, \quad (5)$$

and therefore,

$$\boxed{\nabla \times \vec{E} = 0.} \quad (6)$$

Thus, an electrostatic field is an **irrotational field**. This is the differential form of the statement in (1).

Physical meaning: If a field can be written as the gradient of a scalar potential, then it cannot possess any local “circulation” or “whirling nature”. That is why the curl of electrostatic field is zero.

Circulation Around a Small Rectangular Loop

To understand Stokes' theorem, consider a very small rectangular loop lying in the yz -plane, with sides Δy and Δz . Its area vector is along $\hat{x}$.

The line integral around this closed loop is

$$\begin{aligned} \oint_C \vec{E} \cdot d\vec{l} &= E_y(y, z) \Delta y + E_z(y + \Delta y, z) \Delta z \\ &\quad - E_y(y, z + \Delta z) \Delta y - E_z(y, z) \Delta z. \end{aligned} \quad (7)$$



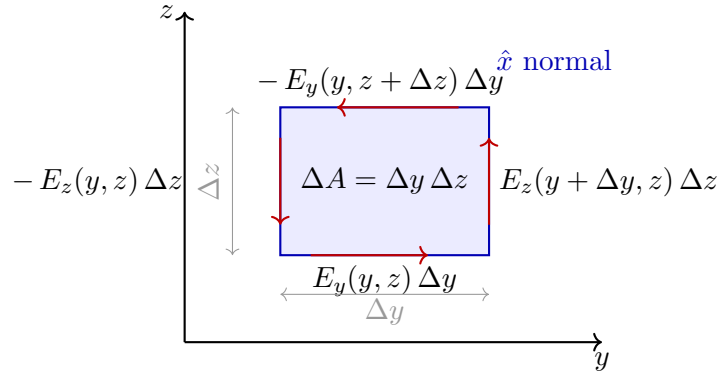


Figure 1: A small rectangular loop in the yz -plane. The circulation around this loop gives the x -component of curl.

Rearranging,

$$\oint_C \vec{E} \cdot d\vec{l} = [E_z(y + \Delta y, z) - E_z(y, z)] \Delta z - [E_y(y, z + \Delta z) - E_y(y, z)] \Delta y. \quad (8)$$

For a very small loop, using first-order expansion,

$$E_z(y + \Delta y, z) - E_z(y, z) \approx \frac{\partial E_z}{\partial y} \Delta y, \quad (9)$$

and

$$E_y(y, z + \Delta z) - E_y(y, z) \approx \frac{\partial E_y}{\partial z} \Delta z. \quad (10)$$

Hence,

$$\begin{aligned} \oint_C \vec{E} \cdot d\vec{l} &= \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) \Delta y \Delta z \\ &= (\nabla \times \vec{E})_x \Delta A. \end{aligned} \quad (11)$$

So, for an infinitesimal surface element,

$$(\nabla \times \vec{E})_x = \lim_{\Delta A \rightarrow 0} \frac{1}{\Delta A} \oint_C \vec{E} \cdot d\vec{l} \quad (12)$$

where the area vector is along $\hat{x}$.

In the same way, if the elemental loop is taken in other coordinate planes, we obtain the other components of curl.

From Small Loops to a Finite Surface

Now consider an arbitrary open surface S bounded by a closed contour C . We can imagine dividing the entire surface into many tiny surface elements.

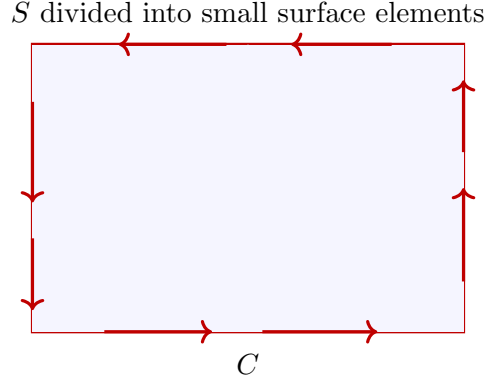


Figure 2: An arbitrary surface subdivided into many small patches. Contributions along common interior edges cancel pairwise, leaving only the circulation along the outer boundary C .

For each small surface element,

$$\oint \vec{E} \cdot d\vec{l} = (\nabla \times \vec{E}) \cdot d\vec{s}. \quad (13)$$

If we sum over all such elemental loops, the contributions along the interior boundaries cancel out because each common edge is traversed once in one direction and once in the opposite direction. Therefore, only the outer boundary survives. Hence,

$$\oint_C \vec{E} \cdot d\vec{l} = \iint_S (\nabla \times \vec{E}) \cdot d\vec{s} \quad (14)$$

This is **Stokes' theorem** for the electric field.

More generally, for any vector field $\vec{A}$,

$$\oint_C \vec{A} \cdot d\vec{l} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{s} \quad (15)$$

where the direction of traversal around C and the direction of $d\vec{s}$ are related by the **right-hand rule**.

Stokes' Theorem in Electromagnetics

The same theorem will be used repeatedly in electromagnetics. For example, for the magnetic flux density $\vec{B}$,

$$\oint_C \vec{B} \cdot d\vec{l} = \iint_S (\nabla \times \vec{B}) \cdot d\vec{s} \quad (16)$$



Similarly, this theorem is the key mathematical bridge between the **integral form** and the **differential form** of Maxwell's equations. In particular, we will use it later while discussing:

- Faraday's law,
- Ampère's law,
- and the rotational nature of electromagnetic fields.

Key Takeaway

$$\oint_C \vec{A} \cdot d\vec{l} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{s}$$

Stokes' theorem states that the **circulation** of a vector field around a closed contour equals the **surface integral of its curl** over any open surface bounded by that contour.

For electrostatics,

$$\nabla \times \vec{E} = 0$$

and hence the electric field has zero circulation around every closed path.

Review of the Curl Operator

To understand **Stokes' theorem** and related theorems more deeply, let us briefly review the meaning of the **curl** operator.

For a vector field $\vec{F}$,

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}. \quad (17)$$

From Stokes' theorem, curl has a very important physical meaning:

$$(\nabla \times \vec{F}) \cdot \hat{n} = \lim_{\Delta S \rightarrow 0} \frac{1}{\Delta S} \oint_C \vec{F} \cdot d\vec{l} \quad (18)$$

Thus, the curl of a vector field measures the **circulation per unit area** in the limit of a very small loop.

So, a field can be called:

- **curl-free** if $\nabla \times \vec{F} = 0$,
- **rotational** if $\nabla \times \vec{F} \neq 0$.

Example 1: A field that changes only along its own direction

Consider the vector field

$$\vec{F} = x \hat{x}. \quad (19)$$

Here, the field points along $\hat{x}$, and its magnitude changes only with x .

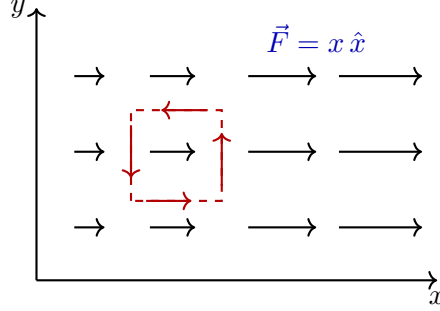


Figure 3: A field whose magnitude changes along the same direction as the field itself. Such a field is curl-free.

Now,

$$\begin{aligned} \nabla \times \vec{F} &= \nabla \times (x \hat{x}) \\ &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & 0 & 0 \end{vmatrix} = 0. \end{aligned} \quad (20)$$

Hence,

$$\boxed{\nabla \times \vec{F} = 0.} \quad (21)$$

Therefore, for a small closed loop,

$$\oint_C \vec{F} \cdot d\vec{l} = 0. \quad (22)$$

Interpretation: Although the field strength changes from left to right, it does not create a local tendency to rotate around a point.

Verification of Example 1 using a small Stokes loop : Consider

$$\vec{F} = x \hat{x}.$$

Since the field has only an x -component, and it depends only on x , the only possibly relevant circulation comes from a small rectangular loop in the xy -plane (to test the z -component of curl).

Take a small rectangular loop with corners

$$(x, y), \quad (x + \Delta x, y), \quad (x + \Delta x, y + \Delta y), \quad (x, y + \Delta y),$$



traversed counterclockwise, so that the area vector is along $\hat{z}$.
Now evaluate the line integral around the loop:

- Bottom edge: $d\vec{l} = dx \hat{x}$, hence

$$\int_{\text{bottom}} \vec{F} \cdot d\vec{l} = \int_x^{x+\Delta x} x' dx'.$$

- Right and left vertical edges: $d\vec{l} \parallel \hat{y}$, whereas $\vec{F} \parallel \hat{x}$, so

$$\vec{F} \cdot d\vec{l} = 0.$$

- Top edge: traversal is from $(x + \Delta x, y + \Delta y)$ to $(x, y + \Delta y)$, hence

$$\int_{\text{top}} \vec{F} \cdot d\vec{l} = - \int_x^{x+\Delta x} x' dx'.$$

Therefore,

$$\oint_C \vec{F} \cdot d\vec{l} = 0.$$

Hence,

$$(\nabla \times \vec{F}) \cdot \hat{z} = \lim_{\Delta x \Delta y \rightarrow 0} \frac{1}{\Delta x \Delta y} \oint_C \vec{F} \cdot d\vec{l} = 0.$$

Also, since $\vec{F}$ has only an x -component and has no variation with y or z in a way that can produce rotation, all components of curl vanish. Thus

$$\boxed{\nabla \times \vec{F} = 0.}$$

So this field is **curl-free**, even though its magnitude changes with position.

Example 2: A field varying perpendicular to its own direction

Now consider

$$\vec{F} = y \hat{x}. \quad (23)$$

The field is still along $\hat{x}$, but now its magnitude changes with y , i.e., in a direction perpendicular to the field direction.

Now,

$$\begin{aligned} \nabla \times \vec{F} &= \nabla \times (y \hat{x}) \\ &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & 0 & 0 \end{vmatrix} \\ &= -\hat{z}. \end{aligned} \quad (24)$$



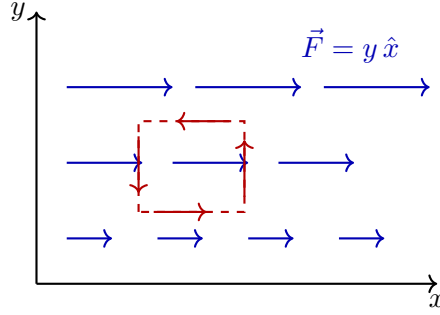


Figure 4: A field that varies in a direction perpendicular to the field direction. This gives nonzero curl.

Hence,

$$\boxed{\nabla \times \vec{F} = -\hat{z} \neq 0.} \quad (25)$$

So the circulation around a small loop is nonzero:

$$\oint_C \vec{F} \cdot d\vec{l} \neq 0. \quad (26)$$

Verification of Example 2 using a small Stokes loop :

Now consider

$$\vec{F} = y \hat{x}.$$

Again, take a small rectangular loop in the xy -plane with area $\Delta S = \Delta x \Delta y$, traversed counterclockwise so that the area vector is along $\hat{z}$.

The circulation is:

- Bottom edge at height y :

$$\int_{\text{bottom}} \vec{F} \cdot d\vec{l} = y \Delta x.$$

- Right and left vertical edges:

$$\vec{F} \cdot d\vec{l} = 0,$$

since $\vec{F} \parallel \hat{x}$ and $d\vec{l} \parallel \hat{y}$.

- Top edge at height $y + \Delta y$, traversed in the negative x -direction:

$$\int_{\text{top}} \vec{F} \cdot d\vec{l} = -(y + \Delta y) \Delta x.$$

Therefore,

$$\oint_C \vec{F} \cdot d\vec{l} = y \Delta x - (y + \Delta y) \Delta x = -\Delta y \Delta x.$$

Dividing by the loop area,

$$\frac{1}{\Delta S} \oint_C \vec{F} \cdot d\vec{l} = \frac{-\Delta x \Delta y}{\Delta x \Delta y} = -1.$$

Hence,

$$(\nabla \times \vec{F}) \cdot \hat{z} = -1,$$

so that

$$\boxed{\nabla \times \vec{F} = -\hat{z}.$$

Thus this field has **nonzero curl**. The key reason is that the field varies in the y -direction, which is perpendicular to the field direction $\hat{x}$.

Important Observation

For the curl to be nonzero, the vector field must vary in a direction **perpendicular** to the field direction. A mere change in magnitude along the same direction of the field does not automatically produce curl.

Circulating Fields in Cylindrical Coordinates

At first glance, one may think that every field with circular field lines must have nonzero curl. That is **not always true**. Curl measures **local circulation density**, not simply whether the field lines look circular. Let us see two important examples in cylindrical coordinates.

For a field of the form

$$\vec{A} = A_\phi(\rho) \hat{\phi}, \quad (27)$$

the z -component of curl is

$$\left(\nabla \times \vec{A} \right)_z = \frac{1}{\rho} \frac{d}{d\rho} (\rho A_\phi). \quad (28)$$

Example 3: Circular field with nonzero curl

Consider

$$\vec{A} = \hat{\phi}. \quad (29)$$

Using (28),

$$\left(\nabla \times \vec{A} \right)_z = \frac{1}{\rho} \frac{d}{d\rho} (\rho) = \frac{1}{\rho}. \quad (30)$$

Hence,

$$\boxed{\nabla \times \vec{A} = \frac{1}{\rho} \hat{z} \neq 0.} \quad (31)$$

So this circular field is **not** curl-free.

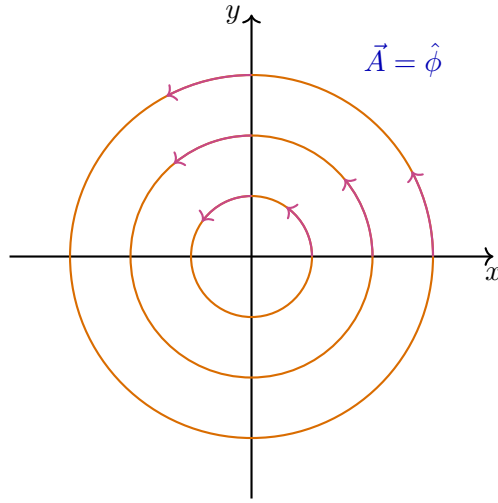


Figure 5: A circulating field with nonzero curl.

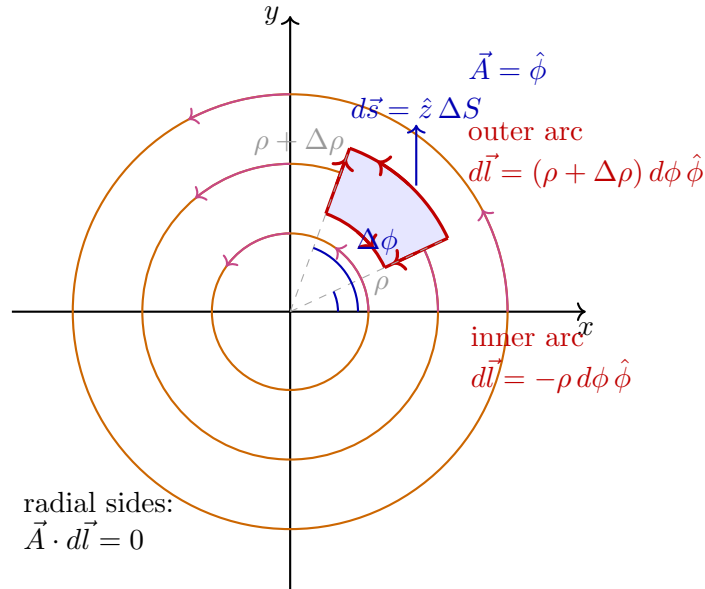


Figure 6: Stokes loop for the field $\vec{A} = \hat{\phi}$. Since the field magnitude is the same on both arcs, but the outer arc is longer than the inner arc, the circulation around the small loop is nonzero.

Verification of Example 3 using a small Stokes loop in cylindrical coordinates Consider

$$\vec{A} = \hat{\phi}.$$

To test curl, choose a small *annular sector* in the $\rho\phi$ -plane, bounded by

$$\rho, \quad \rho + \Delta\rho, \quad \phi, \quad \phi + \Delta\phi,$$



with area vector along $\hat{z}$.

Its small area is approximately

$$\Delta S \approx \rho \Delta \rho \Delta \phi.$$

Now evaluate the line integral around this small loop.

- On the two radial sides, $d\vec{l} \parallel \hat{\rho}$, while $\vec{A} \parallel \hat{\phi}$, so

$$\vec{A} \cdot d\vec{l} = 0.$$

- On the outer arc at radius $\rho + \Delta \rho$,

$$d\vec{l} = (\rho + \Delta \rho) d\phi \hat{\phi},$$

hence

$$\int_{\text{outer arc}} \vec{A} \cdot d\vec{l} = \int_{\phi}^{\phi+\Delta\phi} (\rho + \Delta\rho) d\phi = (\rho + \Delta\rho)\Delta\phi.$$

- On the inner arc at radius ρ , the traversal is in the negative $\hat{\phi}$ direction, so

$$\int_{\text{inner arc}} \vec{A} \cdot d\vec{l} = -\rho \Delta\phi.$$

Thus,

$$\oint_C \vec{A} \cdot d\vec{l} = (\rho + \Delta\rho)\Delta\phi - \rho\Delta\phi = \Delta\rho \Delta\phi.$$

Therefore,

$$\frac{1}{\Delta S} \oint_C \vec{A} \cdot d\vec{l} = \frac{\Delta\rho \Delta\phi}{\rho \Delta\rho \Delta\phi} = \frac{1}{\rho}.$$

Hence,

$$(\nabla \times \vec{A}) \cdot \hat{z} = \frac{1}{\rho},$$

or

$$\boxed{\nabla \times \vec{A} = \frac{1}{\rho} \hat{z}.$$

So, although the field is purely circulating, it indeed has **nonzero local circulation per unit area**, and therefore nonzero curl.

Example 4: Circular field that is locally curl-free

Now consider

$$\vec{A} = \frac{1}{\rho} \hat{\phi}. \quad (32)$$

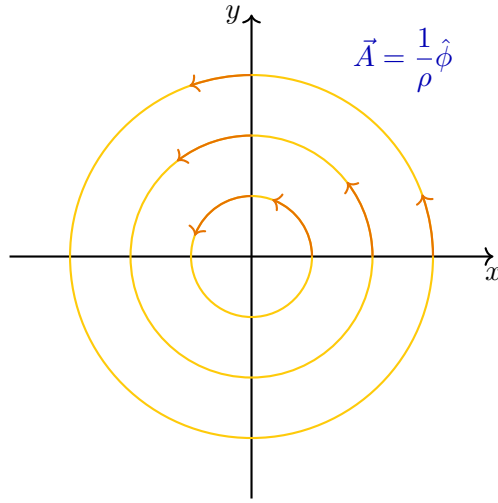


Figure 7: A circular field that is locally curl-free for $\rho \neq 0$.

Again using (28),

$$\left(\nabla \times \vec{A}\right)_z = \frac{1}{\rho} \frac{d}{d\rho} \left(\rho \cdot \frac{1}{\rho} \right) = \frac{1}{\rho} \frac{d}{d\rho} (1) = 0. \quad (33)$$

Therefore,

$$\boxed{\nabla \times \vec{A} = 0 \quad (\rho \neq 0).} \quad (34)$$

So even though the field lines are circular, the field can still be **curl-free** in the region where it is defined.

Verification of Example 4 using a small Stokes loop in cylindrical coordinates :-

Now consider

$$\vec{A} = \frac{1}{\rho} \hat{\phi}.$$

Again choose the same small annular sector:

$$\rho, \quad \rho + \Delta\rho, \quad \phi, \quad \phi + \Delta\phi,$$

with area

$$\Delta S \approx \rho \Delta\rho \Delta\phi.$$

As before, the radial sides contribute zero because

$$\vec{A} \cdot d\vec{l} = 0 \quad \text{on radial segments.}$$

For the two arcs:

- Outer arc at radius $\rho + \Delta\rho$:

$$\int_{\text{outer arc}} \vec{A} \cdot d\vec{l} = \frac{1}{\rho + \Delta\rho} (\rho + \Delta\rho) \Delta\phi = \Delta\phi.$$



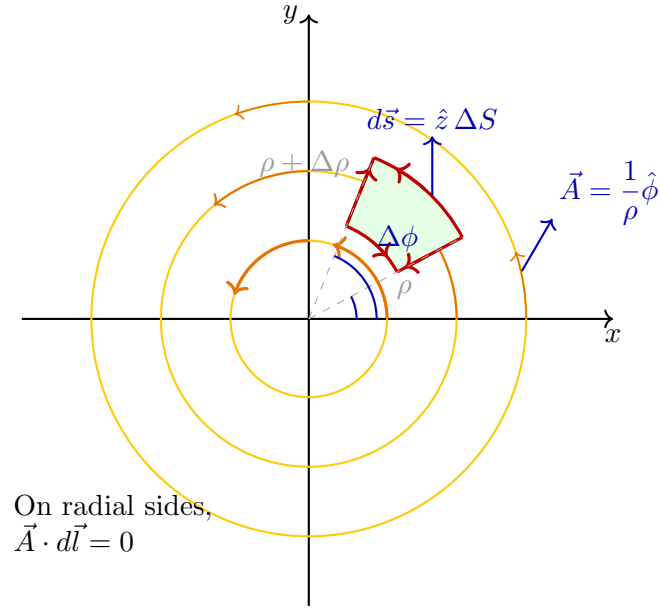


Figure 8: Stokes loop for the field $\vec{A} = \frac{1}{\rho} \hat{\phi}$. The field becomes weaker as ρ increases. The inner arc has stronger field but shorter path, while the outer arc has weaker field but longer path; these effects exactly cancel, so the net circulation around the small loop is zero for $\rho \neq 0$.

- Inner arc at radius ρ , traversed in the negative $\hat{\phi}$ direction:

$$\int_{\text{inner arc}} \vec{A} \cdot d\vec{l} = -\frac{1}{\rho} \rho \Delta\phi = -\Delta\phi.$$

Hence,

$$\oint_C \vec{A} \cdot d\vec{l} = \Delta\phi - \Delta\phi = 0.$$

Therefore,

$$\frac{1}{\Delta S} \oint_C \vec{A} \cdot d\vec{l} = 0,$$

so that

$$(\nabla \times \vec{A}) \cdot \hat{z} = 0.$$

Thus,

$$\boxed{\nabla \times \vec{A} = 0 \quad (\rho \neq 0).}$$

So this field is **locally curl-free**, even though the field lines are circular.

Important caution: If one takes a *finite closed curve* that encloses the origin, then

$$\oint_C \vec{A} \cdot d\vec{l} = 2\pi \neq 0.$$

This does *not* contradict Stokes' theorem, because the field

$$\vec{A} = \frac{1}{\rho} \hat{\phi}$$

is singular at $\rho = 0$, and hence not defined on the entire surface enclosed by such a curve. Therefore, the above local curl-free result is valid only in a region that does **not** include the origin.

Key Concept

Every circulating field does **not** necessarily have nonzero curl.

Curl tells us about **local circulation per unit area**, not merely whether the field lines look curved or circular.

Connection Back to Stokes' Theorem

This review is important because Stokes' theorem connects the **closed line integral** of a vector field around a boundary to the **surface integral of curl** over the enclosed surface:

$$\oint_C \vec{F} \cdot d\vec{l} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{s} \quad (35)$$

Thus:

- if $\nabla \times \vec{F} = 0$, the field is locally curl-free,
- if $\nabla \times \vec{F} \neq 0$, the field has local rotation,
- and Stokes' theorem provides the bridge between the two viewpoints.

In Summary

- Curl measures the **circulation per unit area** of a vector field.

- From Stokes' theorem,

$$(\nabla \times \vec{F}) \cdot \hat{n} = \lim_{\Delta S \rightarrow 0} \frac{1}{\Delta S} \oint_C \vec{F} \cdot d\vec{l}.$$

- For $\vec{F} = x\hat{x}$,

$$\nabla \times \vec{F} = 0,$$

so the field is curl-free.

- For $\vec{F} = y\hat{x}$,

$$\nabla \times \vec{F} = -\hat{z} \neq 0,$$

so the field has nonzero curl.

- A field can have nonzero curl only if there is suitable variation **perpendicular** to the field direction.

- In cylindrical coordinates:

$$\vec{A} = \hat{\phi} \Rightarrow \nabla \times \vec{A} = \frac{1}{\rho} \hat{z} \neq 0,$$

but

$$\vec{A} = \frac{1}{\rho} \hat{\phi} \Rightarrow \nabla \times \vec{A} = 0 \quad (\rho \neq 0).$$

- Therefore, **every circulating field need not always have nonzero curl.**

What is $\nabla \times \vec{B}$?

Let us now ask an important question:

What does $\nabla \times \vec{B}$ represent?

(36)

To understand this, consider the magnetic field produced by a long straight wire carrying steady current I along the z -direction.

Magnetic field due to a straight line current

From Biot–Savart law, for an infinitely long straight current carrying conductor,

$$\vec{B} = \hat{\phi} \frac{\mu_0 I}{2\pi\rho}. \quad (37)$$

Thus the magnetic field lines are concentric circles around the wire.



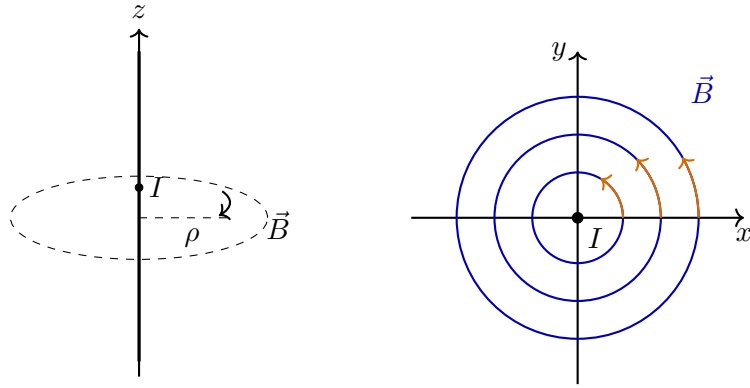


Figure 9: Magnetic field due to a long straight current carrying conductor. The field circulates around the wire.

Line integral around a circular loop

Now choose a circular path of radius ρ centered on the wire.
For this path,

$$d\vec{l} = \rho d\phi \hat{\phi}. \quad (38)$$

Using (37),

$$\begin{aligned} \oint_C \vec{B} \cdot d\vec{l} &= \oint_C \left(\hat{\phi} \frac{\mu_0 I}{2\pi\rho} \right) \cdot (\rho d\phi \hat{\phi}) \\ &= \frac{\mu_0 I}{2\pi} \oint_C d\phi \\ &= \frac{\mu_0 I}{2\pi} \int_0^{2\pi} d\phi \\ &= \mu_0 I. \end{aligned} \quad (39)$$

Hence,

$$\boxed{\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I.} \quad (40)$$

This is the integral result for a circular Amperian loop enclosing current I .

General form: arbitrary closed loop

More generally, if a closed loop encloses several currents, then the total enclosed current is

$$I_{\text{enc}} = \sum_k I_k. \quad (41)$$

So Ampère's law in integral form becomes

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}. \quad (42)$$

If current density $\vec{J}$ is used, then

$$I_{\text{enc}} = \iint_S \vec{J} \cdot d\vec{s}, \quad (43)$$

and therefore

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 \iint_S \vec{J} \cdot d\vec{s}. \quad (44)$$

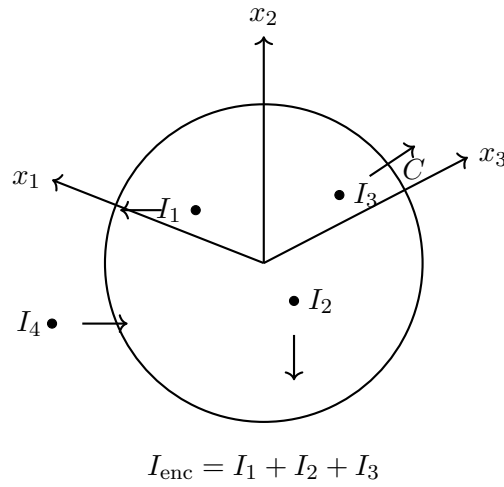


Figure 10: For an arbitrary closed loop, only the current threading the chosen surface contributes to I_{enc} .

Using Stokes' theorem

Now apply Stokes' theorem:

$$\oint_C \vec{B} \cdot d\vec{l} = \iint_S (\nabla \times \vec{B}) \cdot d\vec{s}. \quad (45)$$

Comparing this with (44),

$$\iint_S (\nabla \times \vec{B}) \cdot d\vec{s} = \mu_0 \iint_S \vec{J} \cdot d\vec{s}. \quad (46)$$

Since this must hold for any arbitrary surface,

$$\nabla \times \vec{B} = \mu_0 \vec{J}. \quad (47)$$

This is Ampère's law in differential or point form.



Important Subtlety: Why this does not contradict our earlier curl review

At this point, one may raise a very important question.
From our earlier curl review, we studied the field

$$\vec{A} = \frac{1}{\rho} \hat{\phi},$$

and found that

$$\nabla \times \vec{A} = 0 \quad (\rho \neq 0).$$

But here, for the straight wire,

$$\vec{B} = \frac{\mu_0 I}{2\pi\rho} \hat{\phi},$$

which has the same functional form. Then why are we now writing

$$\nabla \times \vec{B} = \mu_0 \vec{J}?$$

The answer is:

Careful Remark

For the field

$$\vec{B} = \frac{\mu_0 I}{2\pi\rho} \hat{\phi},$$

the curl is indeed

$$\nabla \times \vec{B} = 0 \quad \text{for } \rho > 0,$$

that is, everywhere away from the wire.

However, the current actually flows along the wire axis at $\rho = 0$, and the field is singular there. So the ordinary local calculation of curl applies only in the region where the field is well behaved, namely $\rho \neq 0$.

Thus:

- **away from the wire:**

$$\vec{J} = 0 \quad \Rightarrow \quad \nabla \times \vec{B} = 0,$$

which is fully consistent with our earlier curl example;

- **on the wire axis:** current is present, and this is precisely the location where the magnetic field has its source.

So the nonzero circulation

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

arises only when the chosen loop encloses the current-carrying singular line.
In other words, there is **no contradiction**:



$$\nabla \times \vec{B} = 0 \quad \text{for } \rho > 0, \quad \text{but globally } \oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}} \text{ if the loop encloses the wire.} \quad (48)$$

For a more advanced treatment, one may write the source at the axis using a delta function, but for our present discussion it is enough to remember that the curl is zero everywhere except at the location of the current itself.

Summary

- For a long straight wire carrying current I ,

$$\vec{B} = \frac{\mu_0 I}{2\pi\rho} \hat{\phi}.$$

- The circulation of $\vec{B}$ around a closed loop enclosing the wire is

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}.$$

- Since

$$I_{\text{enc}} = \iint_S \vec{J} \cdot d\vec{s},$$

Ampère's law in integral form becomes

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 \iint_S \vec{J} \cdot d\vec{s}.$$

- By Stokes' theorem,

$$\nabla \times \vec{B} = \mu_0 \vec{J}.$$

- This does **not** contradict the earlier curl example: for the $1/\rho$ type circular field,

$$\nabla \times \vec{B} = 0 \quad \text{for } \rho > 0,$$

i.e. everywhere away from the wire.

- The nonzero closed-loop circulation appears because the loop encloses the current concentrated along the wire axis.

Deriving $\nabla \times \vec{B}$ from the general Biot–Savart law

So far we considered the magnetic field due to a highly symmetric current distribution such as a long straight wire. Now let us take the **general case** and ask:



$$\boxed{\text{Starting from Biot-Savart law, what is } \nabla \times \vec{B} ?} \quad (49)$$

We will do this for a **steady current distribution**.

General Biot-Savart law

Let the observation point be $\vec{r}$ and the source point be $\vec{r}'$. Define

$$\vec{R} = \vec{r} - \vec{r}', \quad R = |\vec{R}|. \quad (50)$$

Then the magnetic field due to a volume current density $\vec{J}(\vec{r}')$ is

$$\boxed{\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \iiint_V \frac{\vec{J}(\vec{r}') \times \vec{R}}{R^3} dv'} \quad (51)$$

where the integration is over the source volume.

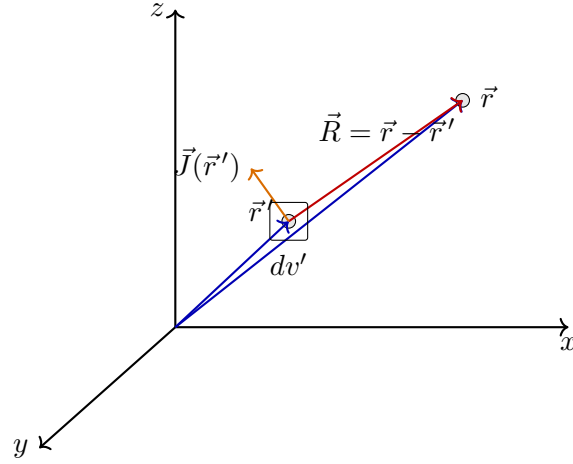


Figure 11: Observation point $\vec{r}$, source point $\vec{r}'$, and the separation vector $\vec{R} = \vec{r} - \vec{r}'$ used in the general Biot-Savart law.

Taking curl of the Biot-Savart law

Now take curl with respect to the observation coordinates $\vec{r}$:

$$\nabla \times \vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \iiint_V \nabla \times \left(\frac{\vec{J}(\vec{r}') \times \vec{R}}{R^3} \right) dv'. \quad (52)$$

It is important to note that here the operator ∇ acts with respect to $\vec{r}$, *not* with respect to $\vec{r}'$. Therefore, $\vec{J}(\vec{r}')$ behaves like a constant vector as far as ∇ is concerned.



To evaluate the curl, use the vector identity

$$\nabla \times (\vec{A} \times \vec{C}) = \vec{A}(\nabla \cdot \vec{C}) - \vec{C}(\nabla \cdot \vec{A}) + (\vec{C} \cdot \nabla)\vec{A} - (\vec{A} \cdot \nabla)\vec{C}. \quad (53)$$

Here we choose

$$\vec{A} = \vec{J}(\vec{r}'), \quad \vec{C} = \frac{\vec{R}}{R^3}.$$

Since $\vec{J}(\vec{r}')$ depends only on the source coordinates, we have

$$\nabla \cdot \vec{J}(\vec{r}') = 0, \quad \left(\frac{\vec{R}}{R^3} \cdot \nabla \right) \vec{J}(\vec{r}') = 0. \quad (54)$$

Hence,

$$\nabla \times \left(\vec{J}(\vec{r}') \times \frac{\vec{R}}{R^3} \right) = \vec{J}(\vec{r}') \left(\nabla \cdot \frac{\vec{R}}{R^3} \right) - (\vec{J}(\vec{r}') \cdot \nabla) \frac{\vec{R}}{R^3}. \quad (55)$$

Substituting (55) into (52), we get

$$\begin{aligned} \nabla \times \vec{B}(\vec{r}) &= \frac{\mu_0}{4\pi} \iiint_V \vec{J}(\vec{r}') \left(\nabla \cdot \frac{\vec{R}}{R^3} \right) dv' \\ &\quad - \frac{\mu_0}{4\pi} \iiint_V (\vec{J}(\vec{r}') \cdot \nabla) \frac{\vec{R}}{R^3} dv'. \end{aligned} \quad (56)$$

Using the required identities

Now we use two important results.

First,

$$\boxed{\nabla \cdot \frac{\vec{R}}{R^3} = 4\pi\delta^{(3)}(\vec{R})} \quad (57)$$

which we will prove in the coming sections.

Second, for a localized steady current distribution, the second volume integral in (56) vanishes. We will also justify this carefully in the coming sections.

Therefore, using (57), (56) reduces to

$$\begin{aligned} \nabla \times \vec{B}(\vec{r}) &= \frac{\mu_0}{4\pi} \iiint_V \vec{J}(\vec{r}') 4\pi\delta^{(3)}(\vec{R}) dv' \\ &= \mu_0 \iiint_V \vec{J}(\vec{r}')\delta^{(3)}(\vec{r} - \vec{r}') dv'. \end{aligned} \quad (58)$$

By the sifting property of the delta function,

$$\iiint_V \vec{J}(\vec{r}')\delta^{(3)}(\vec{r} - \vec{r}') dv' = \vec{J}(\vec{r}). \quad (59)$$



Hence we finally obtain

$$\nabla \times \vec{B} = \mu_0 \vec{J}. \quad (60)$$

This is Ampère's law in differential or point form for magnetostatics.

Result obtained from the general Biot–Savart law

For a steady current distribution,

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \iiint_V \frac{\vec{J}(\vec{r}') \times \vec{R}}{R^3} dv'$$

and taking curl gives

$$\nabla \times \vec{B} = \mu_0 \vec{J}.$$

In this derivation, we used:

$$\nabla \cdot \frac{\vec{R}}{R^3} = 4\pi\delta^{(3)}(\vec{R}),$$

and the fact that the remaining term in the volume integral vanishes for a localized steady current distribution.

Both of these points will be proved in the coming sections.

Proof of the identity $\nabla \cdot \left(\frac{\vec{R}}{R^3} \right) = 4\pi\delta^{(3)}(\vec{R})$

We now prove the first identity used in the derivation of $\nabla \times \vec{B}$ from the general Biot–Savart law:

$$\nabla \cdot \left(\frac{\vec{R}}{R^3} \right) = 4\pi\delta^{(3)}(\vec{R}) \quad (61)$$

This is mathematically the same structure that already appeared in electrostatics for the field of a point charge.

Step 1: Consider the radial vector field

Let us first consider the simpler field

$$\vec{V} = \frac{\hat{r}}{r^2} = \frac{\vec{r}}{r^3}. \quad (62)$$

Later, we will directly replace $\vec{r}$ by $\vec{R} = \vec{r} - \vec{r}'$.

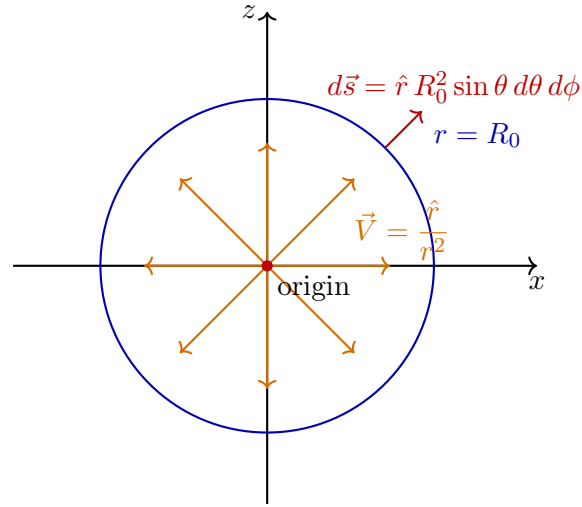


Figure 12: Radial inverse-square field $\vec{V} = \hat{r}/r^2$ and a spherical Gaussian surface of radius R_0 centered at the origin.

Step 2: Evaluate the divergence away from the origin

In spherical coordinates, for a purely radial field

$$\vec{V} = V_r \hat{r}, \quad V_r = \frac{1}{r^2},$$

the divergence is

$$\nabla \cdot \vec{V} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 V_r). \quad (63)$$

Substituting $V_r = 1/r^2$,

$$\begin{aligned} \nabla \cdot \vec{V} &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \cdot \frac{1}{r^2} \right) \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} (1) = 0, \quad r \neq 0. \end{aligned} \quad (64)$$

Thus,

$$\boxed{\nabla \cdot \left(\frac{\hat{r}}{r^2} \right) = 0 \quad \text{for } r \neq 0.} \quad (65)$$

So everywhere away from the origin, the field has zero divergence.

Step 3: Evaluate the flux through a sphere enclosing the origin

Now consider a spherical surface of radius R_0 centered at the origin.



For this sphere,

$$d\vec{s} = \hat{r} R_0^2 \sin \theta d\theta d\phi. \quad (66)$$

Hence,

$$\begin{aligned} \oint_S \vec{V} \cdot d\vec{s} &= \oint_S \left(\frac{\hat{r}}{R_0^2} \right) \cdot (\hat{r} R_0^2 \sin \theta d\theta d\phi) \\ &= \int_0^{2\pi} \int_0^\pi \sin \theta d\theta d\phi \\ &= \left(\int_0^{2\pi} d\phi \right) \left(\int_0^\pi \sin \theta d\theta \right) \\ &= (2\pi)(2) = 4\pi. \end{aligned} \quad (67)$$

Therefore,

$$\boxed{\oint_S \frac{\hat{r}}{r^2} \cdot d\vec{s} = 4\pi.} \quad (68)$$

Step 4: Where is this flux coming from?

Now comes the subtle point.

From (65), we found

$$\nabla \cdot \vec{V} = 0 \quad \text{for } r \neq 0,$$

but from (68), the total outward flux through any sphere enclosing the origin is

$$\oint_S \vec{V} \cdot d\vec{s} = 4\pi.$$

Does this contradict the divergence theorem? No. The reason is that the field $\vec{V} = \hat{r}/r^2$ is singular at the origin. So the ordinary pointwise result $\nabla \cdot \vec{V} = 0$ is valid only for $r \neq 0$.

The missing divergence is concentrated entirely at the origin, and this is represented using the three-dimensional delta function.

Thus, in the generalized sense,

$$\boxed{\nabla \cdot \left(\frac{\hat{r}}{r^2} \right) = 4\pi \delta^{(3)}(\vec{r}).} \quad (69)$$

Indeed, integrating (69) over any volume V gives:

$$\iiint_V \nabla \cdot \left(\frac{\hat{r}}{r^2} \right) dv = \iiint_V 4\pi \delta^{(3)}(\vec{r}) dv = \begin{cases} 4\pi, & \text{if the origin lies inside } V, \\ 0, & \text{otherwise.} \end{cases} \quad (70)$$

That matches exactly the flux behavior of the field.

Step 5: Writing the identity in terms of $\vec{R}$

Now recall that in the Biot–Savart derivation we used

$$\vec{R} = \vec{r} - \vec{r}'.$$

So the above result can be written directly as

$$\nabla \cdot \left(\frac{\vec{R}}{R^3} \right) = 4\pi\delta^{(3)}(\vec{R}) = 4\pi\delta^{(3)}(\vec{r} - \vec{r}'). \quad (71)$$

This is precisely the identity used in deriving

$$\nabla \times \vec{B} = \mu_0 \vec{J}.$$

Boxed Summary

- For the inverse-square radial field

$$\vec{V} = \frac{\hat{r}}{r^2} = \frac{\vec{r}}{r^3},$$

we get

$$\nabla \cdot \vec{V} = 0 \quad \text{for } r \neq 0.$$

- However, the flux through any sphere enclosing the origin is

$$\oiint_S \vec{V} \cdot d\vec{s} = 4\pi.$$

- Therefore, the divergence is not zero everywhere; instead it is concentrated at the singular point $r = 0$.
- This is represented by the 3-D delta function:

$$\nabla \cdot \left(\frac{\hat{r}}{r^2} \right) = 4\pi\delta^{(3)}(\vec{r})$$

- Equivalently,

$$\nabla \cdot \left(\frac{\vec{R}}{R^3} \right) = 4\pi\delta^{(3)}(\vec{R}).$$

Proof that the remaining volume integral is zero

We now prove the second result used in the derivation of $\nabla \times \vec{B}$, namely that

$$\iiint_V \left(\vec{J}(\vec{r}') \cdot \nabla \right) \left(\frac{\vec{R}}{R^3} \right) dv' = \vec{0} \quad (72)$$



for a **localized steady current distribution**.

Here,

$$\vec{R} = \vec{r} - \vec{r}'.$$

This is the term that appeared in the earlier derivation after expanding

$$\nabla \times \left(\vec{J}(\vec{r}') \times \frac{\vec{R}}{R^3} \right).$$

Step 1: Write the integral to be evaluated

Define

$$\vec{I}(\vec{r}) = \iiint_V \left(\vec{J}(\vec{r}') \cdot \nabla \right) \left(\frac{\vec{R}}{R^3} \right) dv'. \quad (73)$$

We want to show that

$$\vec{I}(\vec{r}) = \vec{0}.$$

Now note carefully that ∇ acts with respect to the observation coordinates $\vec{r}$, whereas $\vec{J}(\vec{r}')$ depends only on source coordinates $\vec{r}'$.

Since

$$\vec{R} = \vec{r} - \vec{r}',$$

we have the important relation

$$\nabla \left(\frac{\vec{R}}{R^3} \right) = -\nabla' \left(\frac{\vec{R}}{R^3} \right), \quad (74)$$

because differentiation with respect to $\vec{r}$ and $\vec{r}'$ gives opposite signs.

Therefore,

$$\left(\vec{J}(\vec{r}') \cdot \nabla \right) \left(\frac{\vec{R}}{R^3} \right) = - \left(\vec{J}(\vec{r}') \cdot \nabla' \right) \left(\frac{\vec{R}}{R^3} \right). \quad (75)$$

Substituting into (73),

$$\vec{I}(\vec{r}) = - \iiint_V \left(\vec{J}(\vec{r}') \cdot \nabla' \right) \left(\frac{\vec{R}}{R^3} \right) dv'. \quad (76)$$

Step 2: Evaluate component by component

Let

$$\vec{A}(\vec{r}, \vec{r}') = \frac{\vec{R}}{R^3}, \quad A_k = \frac{R_k}{R^3}, \quad (77)$$

where $k = x, y, z$ denotes any one Cartesian component.

Then the k -th component of $\vec{I}$ is

$$I_k = - \iiint_V \left[J_x \frac{\partial A_k}{\partial x'} + J_y \frac{\partial A_k}{\partial y'} + J_z \frac{\partial A_k}{\partial z'} \right] dv'. \quad (78)$$



Now use the product rule term by term:

$$J_x \frac{\partial A_k}{\partial x'} = \frac{\partial}{\partial x'} (J_x A_k) - A_k \frac{\partial J_x}{\partial x'}, \quad (79)$$

$$J_y \frac{\partial A_k}{\partial y'} = \frac{\partial}{\partial y'} (J_y A_k) - A_k \frac{\partial J_y}{\partial y'}, \quad (80)$$

$$J_z \frac{\partial A_k}{\partial z'} = \frac{\partial}{\partial z'} (J_z A_k) - A_k \frac{\partial J_z}{\partial z'}. \quad (81)$$

Substituting these into (78), we get

$$\begin{aligned} I_k = & - \iiint_V \left[\frac{\partial}{\partial x'} (J_x A_k) + \frac{\partial}{\partial y'} (J_y A_k) + \frac{\partial}{\partial z'} (J_z A_k) \right] dv' \\ & + \iiint_V A_k \left(\frac{\partial J_x}{\partial x'} + \frac{\partial J_y}{\partial y'} + \frac{\partial J_z}{\partial z'} \right) dv'. \end{aligned} \quad (82)$$

Recognizing the divergence terms,

$$I_k = - \iiint_V \nabla' \cdot (A_k \vec{J}) dv' + \iiint_V A_k (\nabla' \cdot \vec{J}) dv'. \quad (83)$$

Step 3: Use steady-current condition

For steady currents, the continuity equation gives

$$\nabla' \cdot \vec{J} = - \frac{\partial \rho_v}{\partial t} = 0. \quad (84)$$

Hence the second volume integral in (83) vanishes, and we obtain

$$I_k = - \iiint_V \nabla' \cdot (A_k \vec{J}) dv'. \quad (85)$$

Now apply the divergence theorem:

$$\iiint_V \nabla' \cdot (A_k \vec{J}) dv' = \oiint_S A_k \vec{J} \cdot d\vec{s}'. \quad (86)$$

Therefore,

$$I_k = - \oiint_S A_k \vec{J} \cdot d\vec{s}'. \quad (87)$$

Step 4: Why the surface integral vanishes

We now choose the integration volume large enough to enclose the entire current distribution. For a **localized current distribution**, the current density vanishes on the outer boundary:

$$\vec{J} = \vec{0} \quad \text{on } S. \quad (88)$$

Hence,

$$\oiint_S A_k \vec{J} \cdot d\vec{s}' = 0, \quad (89)$$

and so from (87),

$$I_k = 0. \quad (90)$$

Since this is true for each component $k = x, y, z$, we conclude that

$$\boxed{\vec{I}(\vec{r}) = \vec{0}.} \quad (91)$$

That is,

$$\boxed{\iiint_V \left(\vec{J}(\vec{r}') \cdot \nabla \right) \left(\frac{\vec{R}}{R^3} \right) dv' = \vec{0}.} \quad (92)$$

Conclusion

Thus, in the earlier derivation of $\nabla \times \vec{B}$, the only surviving term is the one containing

$$\nabla \cdot \left(\frac{\vec{R}}{R^3} \right) = 4\pi\delta^{(3)}(\vec{R}),$$

and therefore

$$\boxed{\nabla \times \vec{B} = \mu_0 \vec{J}.} \quad (93)$$

Boxed Summary

To prove

$$\iiint_V (\vec{J}(\vec{r}') \cdot \nabla) \left(\frac{\vec{R}}{R^3} \right) dv' = \vec{0},$$

we use the following steps:

- Since $\vec{R} = \vec{r} - \vec{r}'$,

$$\nabla \left(\frac{\vec{R}}{R^3} \right) = -\nabla' \left(\frac{\vec{R}}{R^3} \right).$$

- Rewrite the integrand component-wise and use the product rule.

- Use the steady-current condition

$$\nabla' \cdot \vec{J} = 0.$$

- Convert the remaining volume term into a surface integral using divergence theorem.

- For a localized current distribution, $\vec{J} = 0$ on the outer boundary, so the surface integral vanishes.

- Therefore,

$$\iiint_V (\vec{J}(\vec{r}') \cdot \nabla) \left(\frac{\vec{R}}{R^3} \right) dv' = \vec{0}.$$

Alternate derivation of $\nabla \times \vec{B} = \mu_0 \vec{J}$ using the vector potential

Let us now derive the same result in an alternate way, starting from the magnetic vector potential $\vec{A}$.

In magnetostatics, we define

$$\vec{B} = \nabla \times \vec{A}. \quad (94)$$

Therefore,

$$\nabla \times \vec{B} = \nabla \times (\nabla \times \vec{A}). \quad (95)$$

We now show, step by step, that this again leads to

$$\nabla \times \vec{B} = \mu_0 \vec{J}.$$

Step 1: Expression for the vector potential

For a steady localized current distribution,

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \iiint_V \frac{\vec{J}(\vec{r}')}{R} dv', \quad R = |\vec{r} - \vec{r}'|. \quad (96)$$



Here:

$$\vec{R} = \vec{r} - \vec{r}', \quad R = |\vec{R}|.$$

Note that $\vec{J}(\vec{r}')$ depends only on source coordinates, while the differential operator ∇ acts with respect to the observation coordinates $\vec{r}$.

Step 2: Expand $\nabla \times (\nabla \times \vec{A})$

Use the standard vector identity:

$$\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}. \quad (97)$$

Hence,

$$\nabla \times \vec{B} = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}. \quad (98)$$

So we must evaluate:

1. $\nabla \cdot \vec{A}$,
2. $\nabla^2 \vec{A}$.

Step 3: Show that $\nabla \cdot \vec{A} = 0$

Starting from (96),

$$\begin{aligned} \nabla \cdot \vec{A}(\vec{r}) &= \nabla \cdot \left[\frac{\mu_0}{4\pi} \iiint_V \frac{\vec{J}(\vec{r}')}{R} dv' \right] \\ &= \frac{\mu_0}{4\pi} \iiint_V \nabla \cdot \left(\frac{\vec{J}(\vec{r}')}{R} \right) dv'. \end{aligned} \quad (99)$$

Since $\vec{J}(\vec{r}')$ is independent of $\vec{r}$,

$$\nabla \cdot \left(\frac{\vec{J}(\vec{r}')}{R} \right) = \vec{J}(\vec{r}') \cdot \nabla \left(\frac{1}{R} \right). \quad (100)$$

Therefore,

$$\nabla \cdot \vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \iiint_V \vec{J}(\vec{r}') \cdot \nabla \left(\frac{1}{R} \right) dv'. \quad (101)$$

Now use the relation

$$\nabla \left(\frac{1}{R} \right) = -\nabla' \left(\frac{1}{R} \right), \quad (102)$$

so that

$$\nabla \cdot \vec{A}(\vec{r}) = -\frac{\mu_0}{4\pi} \iiint_V \vec{J}(\vec{r}') \cdot \nabla' \left(\frac{1}{R} \right) dv'. \quad (103)$$

Now apply the product rule:

$$\nabla' \cdot \left(\frac{\vec{J}}{R} \right) = \frac{1}{R} (\nabla' \cdot \vec{J}) + \vec{J} \cdot \nabla' \left(\frac{1}{R} \right). \quad (104)$$

Hence,

$$\vec{J} \cdot \nabla' \left(\frac{1}{R} \right) = \nabla' \cdot \left(\frac{\vec{J}}{R} \right) - \frac{1}{R} (\nabla' \cdot \vec{J}). \quad (105)$$

Substituting into (103),

$$\nabla \cdot \vec{A}(\vec{r}) = -\frac{\mu_0}{4\pi} \iiint_V \nabla' \cdot \left(\frac{\vec{J}}{R} \right) dv' + \frac{\mu_0}{4\pi} \iiint_V \frac{\nabla' \cdot \vec{J}}{R} dv'. \quad (106)$$

Now use two facts for steady localized current:

1. $\nabla' \cdot \vec{J} = 0$ (continuity equation for steady current),
2. $\vec{J} = \vec{0}$ on the outer boundary if the current distribution is localized.

Therefore, the second integral is zero, and the first becomes a surface integral which also vanishes:

$$\iiint_V \nabla' \cdot \left(\frac{\vec{J}}{R} \right) dv' = \oint_S \frac{\vec{J}}{R} \cdot d\vec{s}' = 0. \quad (107)$$

Hence,

$$\boxed{\nabla \cdot \vec{A} = 0.} \quad (108)$$

So $\vec{A}$ is divergence-free for magnetostatics in this form.

Step 4: Evaluate $\nabla^2 \vec{A}$

From (96),

$$\begin{aligned} \nabla^2 \vec{A}(\vec{r}) &= \nabla^2 \left[\frac{\mu_0}{4\pi} \iiint_V \frac{\vec{J}(\vec{r}')}{R} dv' \right] \\ &= \frac{\mu_0}{4\pi} \iiint_V \vec{J}(\vec{r}') \nabla^2 \left(\frac{1}{R} \right) dv'. \end{aligned} \quad (109)$$

Again, this is valid because $\vec{J}(\vec{r}')$ does not depend on $\vec{r}$.

Now,

$$\nabla \left(\frac{1}{R} \right) = -\frac{\vec{R}}{R^3}.$$

Therefore,

$$\nabla^2 \left(\frac{1}{R} \right) = \nabla \cdot \nabla \left(\frac{1}{R} \right) = -\nabla \cdot \left(\frac{\vec{R}}{R^3} \right). \quad (110)$$



But from the identity proved earlier,

$$\nabla \cdot \left(\frac{\vec{R}}{R^3} \right) = 4\pi\delta^{(3)}(\vec{R}). \quad (111)$$

Hence,

$$\nabla^2 \left(\frac{1}{R} \right) = -4\pi\delta^{(3)}(\vec{R}). \quad (112)$$

Substituting (112) into (109),

$$\begin{aligned} \nabla^2 \vec{A}(\vec{r}) &= \frac{\mu_0}{4\pi} \iiint_V \vec{J}(\vec{r}') \left[-4\pi\delta^{(3)}(\vec{R}) \right] dv' \\ &= -\mu_0 \iiint_V \vec{J}(\vec{r}') \delta^{(3)}(\vec{r} - \vec{r}') dv'. \end{aligned} \quad (113)$$

Using the sifting property of the delta function,

$$\nabla^2 \vec{A}(\vec{r}) = -\mu_0 \vec{J}(\vec{r}). \quad (114)$$

Step 5: Substitute back into $\nabla \times \vec{B} = \mu_0 \vec{J}$

Now return to (98):

$$\nabla \times \vec{B} = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}.$$

Using (108),

$$\nabla(\nabla \cdot \vec{A}) = \vec{0},$$

and using (114),

$$-\nabla^2 \vec{A} = -(-\mu_0 \vec{J}) = \mu_0 \vec{J}.$$

Therefore,

$$\nabla \times \vec{B} = \mu_0 \vec{J}. \quad (115)$$

Thus, starting from the vector potential also gives exactly the same magnetostatic point form of Ampère's law.

Alternate derivation summary

Starting with

$$\vec{B} = \nabla \times \vec{A}, \quad \vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \iiint_V \frac{\vec{J}(\vec{r}')}{R} dv',$$

we use

$$\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}.$$

Then:

- For a steady localized current distribution,

$$\nabla \cdot \vec{A} = 0.$$

- Also,

$$\nabla^2 \left(\frac{1}{R} \right) = -4\pi \delta^{(3)}(\vec{R})$$

which gives

$$\nabla^2 \vec{A} = -\mu_0 \vec{J}.$$

- Therefore,

$$\nabla \times \vec{B} = \nabla \times (\nabla \times \vec{A}) = \mu_0 \vec{J}.$$